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Resource use and greenhouse gas emissions from grain-finishing beef cattle in seven Australian feedlots: a life cycle assessment

Stephen Wiedemann[A,D], *Rod Davis*[B], *Eugene McGahan, Caoilinn Murphy and Matthew Redding*[C]

[A]Integrity Ag Services, 36a Highfields Road, Highfields, Qld 4352, Australia.
[B]Ostwald Bros., PO Box 279, Dalby, Qld 4405, Australia.
[C]AgriScience Queensland, Department of Agriculture, Food and Forestry, PO Box 102, Toowoomba, Qld 4350, Australia.
[D]Corresponding author. Email: stephen.wiedemann@integrityag.net.au

Abstract. Grain finishing of cattle has become increasingly common in Australia over the past 30 years. However, interest in the associated environmental impacts and resource use is increasing and requires detailed analysis. In this study we conducted a life cycle assessment (LCA) to investigate impacts of the grain-finishing stage for cattle in seven feedlots in eastern Australia, with a particular focus on the feedlot stage, including the impacts from producing the ration, feedlot operations, transport, and livestock emissions while cattle are in the feedlot (gate-to-gate). The functional unit was 1 kg of liveweight gain (LWG) for the feedlot stage and results are included for the full supply chain (cradle-to-gate), reported per kilogram of liveweight (LW) at the point of slaughter. Three classes of cattle produced for different markets were studied: short-fed domestic market (55–80 days on feed), mid-fed export (108–164 days on feed) and long-fed export (>300 days on feed). In the feedlot stage, mean fresh water consumption was found to vary from 171.9 to 672.6 L/kg LWG and mean stress-weighted water use ranged from 100.9 to 193.2 water stress index eq. L/kg LWG. Irrigation contributed 57–91% of total fresh water consumption with differences mainly related to the availability of irrigation water near the feedlot and the use of irrigated feed inputs in rations. Mean fossil energy demand ranged from 16.5 to 34.2 MJ lower heating values/kg LWG and arable land occupation from 18.7 to 40.5 m^2/kg LWG in the feedlot stage. Mean greenhouse gas (GHG) emissions in the feedlot stage ranged from 4.6 to 9.5 kg CO_2-e/kg LWG (excluding land use and direct land-use change emissions). Emissions were dominated by enteric methane and contributions from the production, transport and milling of feed inputs. Linear regression analysis showed that the feed conversion ratio was able to explain >86% of the variation in GHG intensity and energy demand. The feedlot stage contributed between 26% and 44% of total slaughter weight for the classes of cattle fed, whereas the contribution of this phase to resource use varied from 4% to 96% showing impacts from the finishing phase varied considerably, compared with the breeding and backgrounding. GHG emissions and total land occupation per kilogram of LWG during the grain finishing phase were lower than emissions from breeding and backgrounding, resulting in lower life-time emissions for grain-finished cattle compared with grass finishing.

Additional keywords: agricultural systems, feedlots, global climate change.

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Introduction

Grain finishing has become increasingly prevalent in Australia over the past three decades (Bindon and Jones 2001) as a means of increasing cattle growth rates, allowing younger turnoff and potentially heavier finished weights. Market drivers, such as higher carcass weight and marbling in Australia's international markets have also increased demand for grain-finished cattle. In Australia's variable climate, feedlots also have an important role to enable finishing of cattle in drought conditions when grass and forage supply is low. However, the impact of grain finishing on finite resources and the environment is a topic of growing interest both in Australia and internationally and this must be considered as part of a broader discussion on the advantages and costs of alternative beef finishing systems.

Agriculture generated 15.7% of Australia's total direct greenhouse gas (GHG) emissions in 2013 (Commonwealth of Australia 2015*b*), with beef cattle contributing the largest proportion of these GHG emissions. Agriculture also uses 65–70% of water extracted from the environment in Australia (ABS 2006). Global resources of land suitable for agricultural production are also constrained and in Australia, cropping land represents <4% of land mass whereas the area of native and modified pastures represent 56% (Lesslie and Mewett 2013). Hence, the availability of arable land is a more acute concern than the availability of non-arable rangelands. Considering this, alternative production systems such as grain or grass finishing of beef cattle must take into account the different land types required for production. Life cycle assessment (LCA) is a tool

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particularly suited to multi-impact supply chain analysis and has been previously applied to Australian beef case studies (Peters *et al*. 2010*a*, 2010*b*; Eady *et al*. 2011; Ridoutt *et al*. 2012; Wiedemann *et al*. 2016*a*), as well as wheat (Brock *et al*. 2012), wool and sheep meat (Brock *et al*. 2013), pork (Wiedemann *et al*. 2016*b*), poultry (Wiedemann and McGahan 2011; Wiedemann *et al*. 2017) and lamb (Peters *et al*. 2010*a*; Wiedemann *et al*. 2016*c*) case studies. Wiedemann *et al*. (2015) showed that water use and GHG emissions associated with beef production have declined in Australia over the past three decades. One major contributing factor to the decline in GHG has been the improved growth rate in young cattle and the increase in slaughter weight across the national herd (Wiedemann *et al*. 2015). These increases in productivity result in relatively shorter feeding periods to achieve desirable finishing weights, resulting in a decrease in total emissions over the lifetime of an animal. The most significant change in the industry contributing to these effects has been the large increase in grain finished cattle over this time period. Although several studies have compared grain and grass finishing (Peters *et al*. 2010*a*; Pelletier *et al*. 2010), few studies have provided a detailed analysis of the gate-to-gate grain finishing system specifically.

Key differences between grass finishing and grain finishing in feedlots include lower GHG emissions from enteric methane (Dong *et al*. 2006) and potentially higher emissions from manure management (Commonwealth of Australia 2015*b*) for feedlot-finished cattle. Manufacture of feed inputs requires inputs of fertiliser and machinery operations, increasing fossil fuel energy use and potentially GHG emissions compared with grass finishing. Transport-related effects are also evident. Feedlot operations and feed preparation are also energy intensive processes associated with grain finishing. Fossil fuel and water inputs for grain milling and feedlot operations have been quantified for Australian feedlots by Davis *et al*. (2010*a*, 2010*b*). However, these studies did not include some aspects of the wider feedlot system, such as the production of feed inputs and therefore a comprehensive understanding of water use for grain finishing has not been completed to date. The present study aimed to (i) determine resource use and GHG emissions from Australian beef finished in feedlots, targeting three different market specifications, (ii) to provide benchmarking data and identify the relative impacts of different inputs and processes on GHG and resource use, and (iii) to quantify the impact of different finishing systems on the full cradle-to-gate impacts for three different market types commonly fed in Australia.

Materials and methods

Goal and scope

This study reports a gate-to-gate LCA of seven beef feedlots in eastern Australia. It is a companion study to Wiedemann *et al*. (2016*a*), which used LCA to investigate environmental impacts from grass-finished beef production in eastern Australia to the farm gate. The study investigated global warming (aggregated GHG emissions, including impacts from land use and direct land use change – LU and dLUC). Fossil fuel energy demand was assessed using the Fossil Fuel Depletion indicator (Frischknecht *et al*. 2007), measured in mega-joules (MJ) using lower heating values (LHV). Stress-weighted water use was assessed using the, water stress index (WSI) values from Pfister *et al*. (2009). Results from two life cycle inventory (LCI) methods were also assessed: fresh water consumption and land occupation. Results were presented using a functional unit (FU) of kilograms of liveweight gain (kg LWG) in the grain finishing phase (gate-to-gate) and impacts from cradle-to-gate were assessed using data for feeder cattle reported by Wiedemann *et al*. (2016*a*) and reported per kilogram of liveweight (LW) immediately before meat processing. The gate-to-gate analysis included all processes and inputs associated with grain finishing in commercial feedlots, including the impacts from producing the ration, feedlot operations, transport, and livestock emissions while cattle are in the feedlot.

Description of the case-study feedlots

The seven feedlots (FL) ranged in throughput from 6000 to 130 000 head of cattle annually, with a total throughput for the years 2007 and 2008 of 330 000 comprising ~15% of industry throughput for these years (ALFA and MLA 2007, 2008). Feedlots were located in: central Queensland (Qld – FL 1), southern Qld (FL 2, 3, 4), northern New South Wales (NSW – FL 5), southern NSW (FL 6) and Victoria (Vic. – FL 7). Cattle fed to three different market specifications were investigated: short-fed domestic market (55–80 days on feed), mid-fed export (108–164 days on feed) and long-fed export (>300 days on feed).

Description of supply chains

Average feeder cattle production systems from northern NSW were used to provide impacts associated with breeding and backgrounding cattle before feedlot entry. Data were modelled from the NSW regional average farm dataset described in Wiedemann *et al*. (2016*a*), and differed only with respect to the feedlot entry weight requirements for the different market classes. Although breed and genotype differences may exist between cattle entering different grain-finished markets, data were not available to differentiate between these herds at the farm and backgrounding stages, and differences are not expected to be large. Impacts associated with LU and dLUC in the breeding and backgrounding stage were determined using data from Henry *et al*. (2015).

Life cycle inventory

Detailed production data, livestock inventories and input data were collected from feedlot financial records, interviews and site visits (Davis *et al*. 2010*a*, 2010*b*). Livestock data were reported from feedlot records of the number and weight of cattle entering and leaving the feedlot, and the recorded LWG and dry matter intake (DMI). LWG was corrected for shrink in cattle entering the feedlot to avoid over estimating LWG during the feeding period. Production characteristics are provided in Tables 1 and 2. Fossil fuel energy demand was determined from inventories of purchased fossil fuels and electricity use at the feedlot (Tables 3, 4) and from impacts associated with commodity use. Major energy inputs at the feedlot were mainly related to feed processing (electricity and gas use) and diesel for vehicle operation, including pen cleaning and feed delivery. Transport records for livestock movements,

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Table 1. Production parameters for three market types of feedlot-finished cattle

	Domestic grain short-fed (n = 5)		Export grain mid-fed (n = 4)		Export grain long-fed (n = 1)	
	Mean	Range	Mean	Range	Mean	Range
Entry weight (kg LW)	347	327–370	421	392–463	441	n.a.
Days on feed	69	55–80	125	108–164	346	n.a.
Average daily gain (kg/day)	1.75	1.7–1.9	1.84	1.7–2.0	0.99	n.a.
Feed conversion ratio (kg/kg) – dry matter basis	5.2	4.6–5.7	5.8	5.3–6.4	10.2	n.a.
Exit weight (kg LW)	468	438–502	652	594–730	784	n.a.
Mortality rate	0.8%	0.16–1.65%	0.7%	0.40–1.30%	2.1%	n.a.
Dry matter intake (kg/head.day)	9.0	8.5–9.5	10.7	10–11.9	10.1	n.a.

Table 2. Production parameters for breeding, backgrounding and finishing of grain fed cattle

	Short-fed domestic feedlot finished	Mid-fed export feedlot finished	Long-fed export feedlot finished
	Breeding[A]		
Days on farm	365	365	365
Average daily gain (kg)	0.63	0.63	0.63
Closing liveweight (kg – young cattle)	263	263	263
Closing liveweight (kg – breeding cattle)	125	125	125
	Backgrounding[A]		
Days on farm	153	338	342
Average daily gain	0.55	0.47	0.52
Closing liveweight (kg – backgrounding cattle)	347	421	441
	Finishing		
Days on feed – feedlot	69	125	346
Average daily gain	1.75	1.84	0.99
Closing liveweight (kg – finishing cattle)	468	652	784

[A]Data sourced from NSW regional average farm (RAF) dataset reported in Wiedemann *et al*. (2016*a*).

Table 3. Feedlot purchased energy for general operations (excludes feed-milling), reported per 1000-head days

	FL1	FL2	FL3	FL4	FL5	FL6	FL7
Electricity (kWh/1000-head day[A])	19.55	58.37	63.84	84.73	17.01	19.84	68.83
Diesel (L/1000-head day)	11.34	16.75	12.14	19.50	7.66	13.86	12.67
Gas (MJ/1000-head day)	0.83	0.00	0.00	0.00	0.00	0.00	0.00
Petrol (L/1000-head day)	0.72	0.81	8.16	9.65	11.03	11.76	6.90

[A]Head days represent one animal occupying the feedlot for 1 day.

purchased inputs and staff movements were determined from feedlot inventory data and were included in the analysis. Impacts associated with infrastructure, such as feedlot construction, were excluded based on the findings of a scoping study showing the contribution from these was <1% (S. G. Wiedemann, R. J. Davis and E. J. McGahan, unpubl. data). Impacts associated with services such as communications, insurance and accounting, were modelled based on expenditure, using economic input-output data (Rebitzer *et al*. 2002).

Dry matter intake was determined from records of feed delivery to the feed bunk at each feedlot, converted from an as-fed basis to DMI using recorded or calculated ration moisture contents. Feed intake and feed characteristics were used to model livestock GHG emissions, using methods described below. Background data were sourced from the Australian LCI database (Life Cycle Strategies 2007) where available, or the European Ecoinvent (2.2) database (Swiss Centre for Life Cycle Inventories 2010). Feed grain inventory processes were detailed in Wiedemann *et al*. (2016*b*) and Wiedemann and McGahan (2011). Commodity data, and inputs associated with feed milling, are provided in Tables 4 and 5, respectively.

Fresh water consumption

The water-use inventory was developed using measured water data described in Davis *et al*. (2010*a*); and modelling of dam water supply systems and dam supply efficiencies using

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Table 4. Feed-milling energy and water inputs per tonne of ration milled and delivered to the feed bunk

	FL1	FL2	FL3	FL4	FL5	FL6	FL7
Electricity (kwh/tonne[A])	8.05	6.61	4.70	7.08	6.68	5.54	5.36
Diesel (L/tonne)	0.85	0.83	2.63	2.27	0.62	1.77	1.55
Gas (MJ/tonne)	180.65	128.90	0.00	0.00	163.91	163.21	191.59
Petrol (L/tonne)	0.00	0.00	0.00	0.00	0.03	0.00	0.00
Water (L/tonne)	138.89	130.61	189.36	42.44	194.29	80.46	197.27

[A]Reported on an 'as-fed' basis inclusive of moisture.

Table 5. Commodity purchases per tonne of ration (as-fed) for three market classes, averaged over 2 years

Ration component[A]	Short-fed domestic	Mid-fed export	Long-fed export
Barley (10%)	75.7	94.7	136.7
Maize (8%)	68.7	0.0	0.0
Sorghum (10%)	311.2	289.4	157.3
Wheat (13%)	225.6	295.8	309.3
Canola meal (36%)	0.7	0.9	1.3
Cottonseed meal (38%)	20.1	23.9	0.0
White fluffy cottonseed	61.8	63.4	0.0
Hay[B]	26.1	33.6	2.9
Straw[C]	10.6	3.2	89.0
Silage[D]	111.7	85.1	199.0
Cotton hulls	4.3	17.2	0.0
Oil	7.2	12.2	1.6
Molasses	25.1	21.9	71.2
Dry supplement	0.0	0.0	31.7
Wet supplement	51.2	58.7	0.0
Total (kg)	1000	1000	1000

[A]Protein content in brackets.
[B]Includes lucerne, sorghum and wheat hay.
[C]Includes corn, wheat, triticale and sorghum straw.
[D]Includes corn, sorghum, oaten and wheat silage.

methods outlined in Wiedemann *et al*. (2016*a*). Details of the water supply system and system efficiency are provided in Table 6. Water supply was predominantly from bores located on-site, though dams were also an important water source. All dams either captured overland flow from the local area, or were pumped from local river systems.

Australian feedlots are designed to control drainage and overland flow around the feedlot site to restrict movement of manure nutrients to the environment (Skerman 2000), thus restricting runoff to the environment compared with the reference site before feedlot construction. The change in runoff associated with LUC was included in the fresh water consumption inventory following guidance from ISO (2014) by modelling long-term runoff from each feedlot based on the soil and vegetation properties of the reference site before feedlot construction. Runoff modelling was done using USDA-SCS KII curve numbers (USDA NRCS 2007). Data are shown in Table 7.

Irrigation water use associated with feed inputs was assessed both for ration inputs produced at feedlot farms (described as 'ration irrigation on-site') and that associated with purchased commodities (described as 'ration irrigation off-site'). Supply losses associated with on-site irrigation were also determined from farm water balances following methods described in Wiedemann *et al*. (2016*a*). Methods described by Wiedemann *et al*. (2015) were used to model water use associated with irrigation used in the production of purchased feed. Irrigation water used to produce hay and silage at each feedlot was recorded as part of the survey. Irrigation water use associated with water-intensive inputs such as cotton seed, a co-product of cotton lint production, was modelled using inventory data from Tan *et al*. (2013) and water-use data from ABS (2011). Water-use was allocated between cotton lint and seed on an economic basis, with 10% of total irrigation water use attributed to cotton seed production.

Stress-weighted water use

Stress-weighted water use is a weighted indicator of the volume of water from highly stressed water catchments used by the system. This was determined by multiplying the volume of fresh water used by the system in each region with the appropriate WSI values from Pfister *et al*. (2009). The value was then divided by the global average WSI (0.602) and expressed as a water equivalent (H_2O-e; Ridoutt and Pfister 2010).

Land occupation

Land occupation was determined from records regarding the size of each feedlot and the land occupation associated with feed production. The land footprint of each feedlot (Table 6) was determined using geographic information systems (GIS) software and satellite imagery, whereas land occupation data for feed inputs were derived from inventory datasets. No characterisation factors were applied, and land occupation results were reported in square metre years (m^2 year).

Livestock and manure GHG emissions

Enteric methane was modelled using the regression of Moe and Tyrrell (1979) as applied by the Australian National Inventory Report (NIR) (Commonwealth of Australia 2015*b*). To examine the sensitivity of total feedlot emissions to this factor, a sensitivity analysis was performed by re-calculating enteric methane using IPCC 2006 factors of 3% gross energy intake (GEI) and 6.5% GEI (Dong *et al*. 2006). Manure emission methods were determined using methods outlined in the NIR (Commonwealth of Australia 2015*b*) and are shown in Table 8. Uncertainty estimates for manure emission factors were based on the IPCC (Dong *et al*. 2006). Global warming potentials of 25 for methane and 298 for nitrous oxide (Solomon *et al*. 2007) were applied.

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Table 6. Land and water resources for seven feedlots in eastern Australia

	FL1	FL2	FL3	FL4	FL5	FL6	FL7
Location	Qld	Qld	Qld	Qld	NSW	NSW	Vic.
Average annual rainfall (mm)[A]	555	582	624	662	526	857	430
Land use							
Feedlot area – non-arable (ha)	102.0	47.7	76.0	25.9	71.0	149.0	115.0
Feedlot water supply							
Dam (%)	100%	80%	25%	0%	0%	10%	0%
Bore (%)	0%	20%	75%	100%	2%	0%	100%
Creek (%)	0%	0%	0%	0%	0%	90%	0%
Reticulated (%)	0%	0%	0%	0%	98%	0%	0%
Dam efficiency factor	0.28	0.42	0.78	n.a.	n.a.	0.11	n.a.
Bore efficiency factor	n.a.	0.95	0.95	0.95	0.95	n.a.	0.95
Total water supply (ML)	221.3	305.9	157.0	15.2	293.1	319.5	261.7
Water stress index (L/L)	0.855	0.021	0.021	0.021	0.815	0.021	0.815

[A]Recorded for nearest major town.

Table 7. Runoff from reference land occupation attributed to feedlot cattle production

	FL1	FL2	FL3	FL4	FL5	FL6	FL7
Runoff from reference land occupation (ML/year)	49.0	14.5	24.3	2.8	14.1	23.1	10.4
Fresh water consumption attributed to cattle production (L/head finished)	1069.8	119.9	719.7	389.1	283.5	877.6	232.0

As noted by Redding *et al*. (2015), the NIR method for feed pad nitrous oxide is not substantiated by adequate field research. Redding *et al*. (2015) found feed pad nitrous oxide emissions differed in both the mechanisms influencing emissions and the intensity of emissions compared with the Australian inventory approach. To examine the sensitivity of total feedlot emissions to this factor, we applied a comparison method to predict nitrous oxide from manure deposited to the feed pad, using data from Redding *et al*. (2015) and P. Shorten and M. Redding (unpubl. data). This method used climate-specific nitrous oxide emission factors, based on emissions per m^2 of pen surface area occupied per animal (stocking density) in each feedlot region (see Table 8).

GHG emissions (including dLUC) – feedlot operations and feed inputs

The GHG emissions associated with fossil fuel use at the feedlot were determined from the inventory of fossil fuel use in Davis *et al*. (2010*b*). GHG emissions arising from grain or forage production were determined from the databases described. Previous grain studies reported in Wiedemann *et al*. (2010) and Brock *et al*. (2012) have not included soil carbon losses associated with LU and dLUC and emissions from these sources remain uncertain, though inclusion of these sources is recommended for feed grains (LEAP 2014). Two methods were used to determine potential emissions from these sources. First, we determined total LU and dLUC, emissions from an analysis of data from the Australian national inventory (Commonwealth of Australia 2015*c*). This showed annualised, national emissions associated with conversion of forest land to crop land (dLUC) were 4 755 000 t CO_2-e in the period 1990–2010 (Commonwealth of Australia 2015*a*). National LU emissions from crop land were –4 800 000 t CO_2-e (negative emissions indicate carbon sequestration), annualised over the same period. Carbon sequestration in Australian crop land is mostly in response to carbon sequestration resulting from adoption of improved cropping practices over the past 20 years, as shown by Luo *et al*. (2010). When divided by the average total land area sown to crops annually in Australia over the period 1990–2010, annualised emissions from LU and dLUC were –229 and 227 kg CO_2-e/ha. Alternatively, LU and dLUC emissions were estimated using a method adapted from Wiedemann *et al*. (2016*c*). This accounted for different levels of carbon loss for crop land managed with tillage or zero tillage in each state, and accounted for dLUC as a result of the expansion of crop land in the eastern states of Australia (Lesslie and Mewett 2013; Wiedemann *et al*. 2016*c*). Total reported LU and dLUC emissions applied to crop land using this method were 0.44, 0.51 and 0.67 t CO_2-e/ha.year for NSW, southern NSW/Vic., and Qld, respectively, and a detailed explanation of GHG emissions from LU and dLUC is presented in the supplementary material.

Handling co-production

The feedlot system produced beef and manure. Manure is sold as a fertiliser replacement and soil conditioner and has a very low value compared with beef. Co-production of manure was handled by system expansion taking into account the avoided fertiliser for crop production that would be required in the absence of feedlot manure, using the method described by Wiedemann *et al*. (2010). This method accounted for the nitrogen and phosphorus content in manure, and the relative emissions from manure or avoided fertiliser use, after losses associated with spreading were taken into account.

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Table 8. Greenhouse gas emission factors with uncertainty for modelling livestock emissions from feedlot cattle

Emission source	Emission and units	Key parameters/model	Assumed uncertainty	Reference
Enteric fermentation	kg CH_4/head	3.406 + 0.510SR + 1.736H + 2.648C	SR: 0.23–0.79 H: 1.47–2.00 C: 2.46–2.83	Commonwealth of Australia (2015*b*); Moe and Tyrrell (1979)
Manure – emissions from pen surface	MCF	3% (Qld) 1% (NSW, Vic.)	±20%	Commonwealth of Australia (2015*b*)
	NH_3-N/kg N excreted	0.60	±50%	Commonwealth of Australia (2015*b*);
	N_2O-N/kg N excreted	0.02	Factor of 2	Commonwealth of Australia (2015*b*); Dong *et al*. (2006)
Manure – emissions from stockpile	MCF	2%	±20%	Dong *et al*. (2006)
	NH_3-N/kg N to stockpile	0.25	±25%	Commonwealth of Australia (2015*b*); DEWR (2007) Watts *et al*. (2012)
	N_2O-N/kg N to stockpile	0.005	Factor of 2	Commonwealth of Australia (2015*b*); Dong *et al*. (2006)
Effluent – from anaerobic storage	MCF	77% (Qld) 75% (NSW, Vic.)	±20%	Commonwealth of Australia (2015*b*); Dong *et al*. (2006)
	NH_3-N/kg N to pond	0.35	0.25–0.75	Commonwealth of Australia (2015*b*); Dong *et al*. (2006)
Land application	N_2O-N/kg N to land	0.01	0.003–0.03	De Klein *et al*. (2006); Commonwealth of Australia (2015*b*)
	NH_3-N/kg N to land	0.2	0.18–0.22	Watts *et al*. (2012)
Indirect N_2O from volatilised NH_3	N_2O-N/kg N volatilised as NH_3	0.002	Factor of 2	Commonwealth of Australia (2015*b*)
Sensitivity analyses				
Enteric methane	$CH_{4\%}$ of Gross energy intake	3% – high concentrate diet (domestic and mid-fed), 6.5% low concentrate diet (long fed)	± 1%	Dong *et al*. (2006)
Manure – from pen surface	N_2O/m^2 day	0.039 (Nth Qld) 0.033 (Sth Qld) 0.012 (Nth NSW) 0.016 (Sth NSW) 0.005 (Vic.)	Factor of 2	Redding *et al*. (2015) P. Shorten and M. Redding (unpubl. data)
Stocking density	m^2/head	16.3 (FL1, FL5) 12.2 (FL2) 24.5 (FL3) 12.8 (FL4) 18.2 (FL6) 17.5 (FL7)		Feedlot data

Data analyses

The study was modelled using SimaPro 8 (Pré-Consultants 2014). Model uncertainty for the manure management emissions was assessed using Monte Carlo analysis in SimaPro 8. One-thousand iterations provided a 95% confidence interval for the results. Comparison between the short-fed and mid-fed market types was made using the *t*-test in *R* (R Development Core Team 2014). Results from the single long-fed market type analysis are described qualitatively. Linear regression analysis was used to describe the influence of key production parameters and farm variables on GHG emissions and resources using *R*.

Results

Fresh water consumption and stress-weighted water use – feedlot gate-to-gate

Mean fresh water consumption was dominated by irrigation water use and supply losses. Off-site irrigation and supply losses contributed 35–57% of total fresh water consumption (see Fig. 1), whereas on-site irrigation and supply losses contributed 43–56% (Fig. 1). No correlation was found between fresh water use and feed conversion ratio or average daily gain. Mean fresh water consumption was not significantly different between the short-fed and mid-fed market types (see Table 9). Fresh water consumption was lower from the long fed feedlot because of lower on-site irrigation and the lower quantities of specific, water intensive ration commodities used.

Mean stress-weighted water use was not significantly different between the short-fed and mid-fed cattle (Table 9) and tended to be lower for the long-fed feedlot. Stress-weighted water use was related to both the volume of water used and regional differences in the WSI.

Land occupation – feedlot gate-to-gate

Mean crop land occupation was highest for the long-fed cattle (40.5 m^2/kg LWG), whereas the land occupation for the

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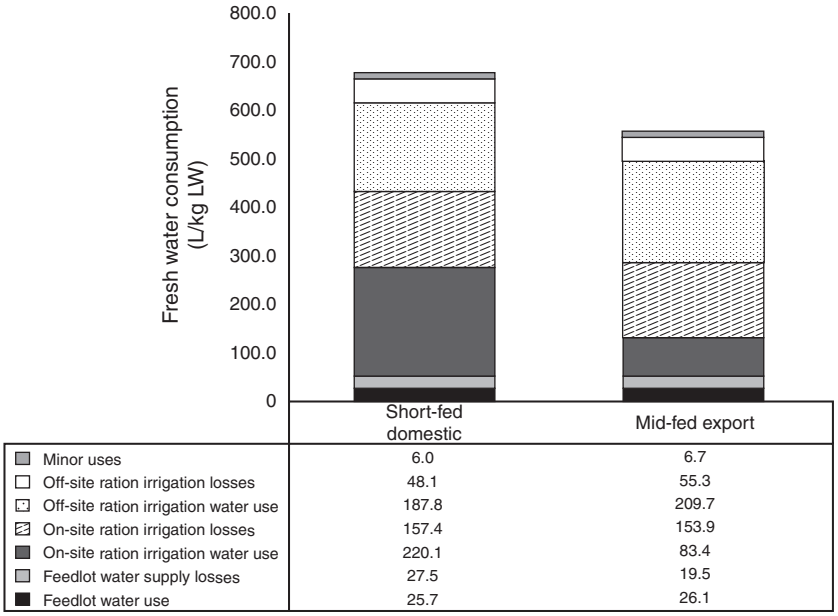

Fig. 1. Contribution to fresh water consumption per kg of liveweight gain from domestic short-fed and mid-fed export feedlot steers.

Table 9. Resource use and environmental impacts per kg of liveweight gain from short-fed, and mid-fed and long-fed export feedlot steers
Different letters indicate significant differences ($P < 0.05$) between short-fed and mid-fed market classes

Market	Impact/inventory categories									
	Fossil energy (MJ lower heating values)		Fresh water consumption (L)		Water stress (eq L)		Crop land (m^2)		Global warming (excluding land use change) (kg CO_2-e)	
	Mean	Range	Mean	Range	Mean	Range	Mean	Range	Mean	Range
Short-fed Domestic (n = 5)	16.5a	15.5–18.1	672.6a	221.1–1246	193.2a	17.9–484	18.7a	17.2–20.9	4.6a	4.3–5.2
Mid-fed export (n = 4)	18.9b	16.3–23.3	554.6a	246.1–1087.3	191.8a	19.1–510.1	22.5b	18.4–28.7	5.2b	4.8–6.1
Long-fed export (n = 1)	34.2	n.a.	171.9	n.a.	100.9	n.a.	40.5	n.a.	9.5	n.a.

short-fed (18.7 m^2/kg LWG) and mid-fed cattle (22.5 m^2/kg LWG) was considerably lower (see Table 9 and Fig. 2).

Energy demand – feedlot gate-to-gate

Mean fossil fuel energy demand was significantly lower ($P < 0.05$) for the short-fed cattle (16.5 MJ LHV/kg LWG) than the mid-fed cattle (18.9 MJ LHV/kg LWG). Energy demand was considerably higher in the long-fed market class (see Table 9 and Fig. 3). Feed production and milling were the largest contributors to energy demand. Among individual feedlots, energy demand varied by 43% between the lowest and highest values for the short-fed and mid-fed market classes. The linear regression of energy demand and feed conversion ratio showed that this parameter explained 0.86 of the variability ($P < 0.001$) according to the following equation:

$$\text{Energy demand} = 3.34 \pm 0.36 \text{ feed conversion ratio} - 0.83 \pm 1.69 \ (R^2 = 0.86)$$

Although the small sample size (n = 1) precluded statistical comparison of the long-fed system with the shorter feeding periods, there was a strong trend towards lower energy impacts from the shorter feeding periods.

GHG emissions – feedlot gate-to-gate

The mean estimated GHG intensity (excluding LU and dLUC) was 4.6 and 5.2 kg CO_2-e/kg LWG for the short-fed and mid-fed cattle with a 30% variation between the highest and lowest values between feedlots. Emissions from short-fed and mid-fed cattle were lower than for long-fed cattle, and were significantly different from each other ($P < 0.05$). The GHG contribution analysis showed mean contribution (excluding LU and dLUC) by gas type was similar across market classes. When analysed by gas type, methane contributed 51% predominantly from enteric sources, CO_2 from fossil fuels contributed 22%, and nitrous oxide from manure management and cropping contributed 27% of total impacts (Fig. 4). When emissions were grouped according to different subsystems, 26% of GHG impacts arose from ration production, transport and milling, 3% from feedlot operations (excluding milling), 23% from manure emission sources, and 48% from livestock enteric methane.

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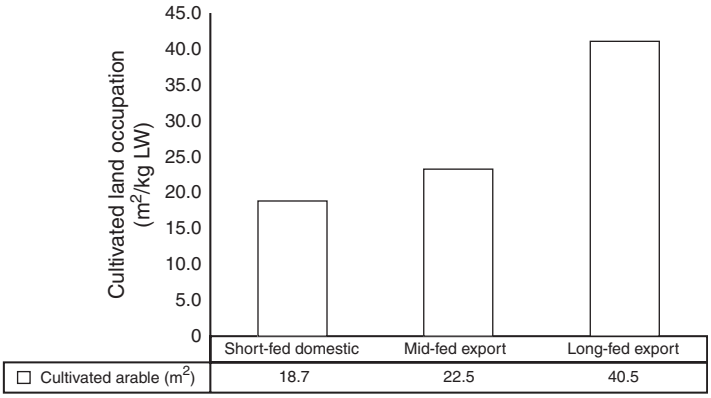

Fig. 2. Arable land occupation per kg of liveweight gain from domestic short-fed, and mid-fed and long-fed export feedlot steers.

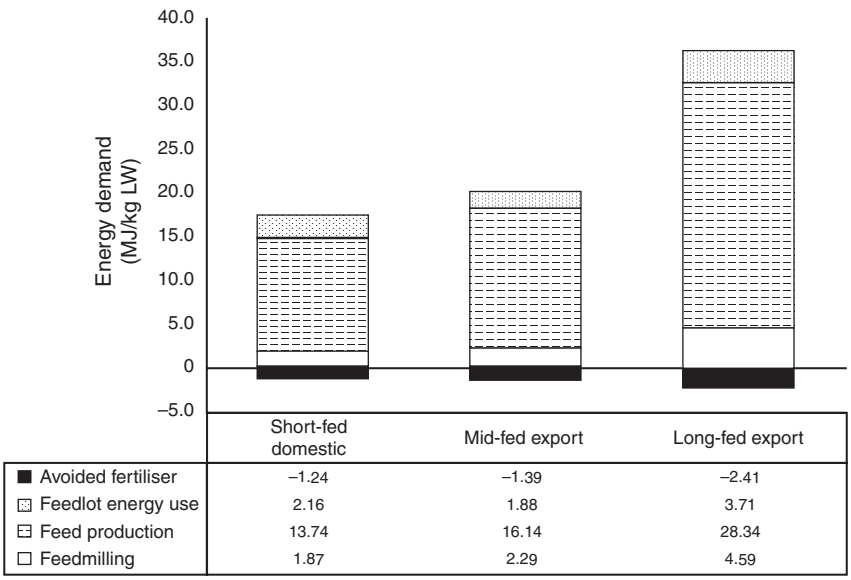

Fig. 3. Energy demand per kg of liveweight gain from domestic short-fed, and mid-fed and long-fed export feedlot steers.

The linear regression analysis showed that feed conversion ratio was able to explain 0.95 of the variation ($P < 0.001$) in GHG intensity. The regression model of GHG emissions intensity was:

$$\text{GHG} = 0.979 \pm 0.06 \text{ feed conversion ratio} - 0.50 \pm 0.36 \ (R^2 = 0.95)$$

Estimated emissions intensity was sensitive to the method chosen for predicting livestock emissions. Applying IPCC defaults for predicting enteric methane revealed ~40% lower enteric methane for the domestic and mid-fed cattle per kg LWG, whereas enteric methane from long-fed cattle increased by 26% with the IPCC default method applied. Results were sensitive to the factors used to predict manure GHG emissions. Applying alternative emission factors for pen nitrous oxide resulted in a decrease in estimated feedpad nitrous oxide from 75.0 to 14.0 kg CO_2-e/100 kg LWG for the southern Qld feedlot, and from 101.1 to 5.3 kg CO_2-e/100 kg LWG for the southern NSW feedlot. This resulted in total manure management emissions decreasing 60% and 81% for the southern Qld and southern NSW feedlots, respectively. The impact on total emissions from all sources (excluding LU and dLUC), when the sensitivity analysis emission factors from Table 8 were used, showed a decrease of 16% for the short-fed market class, 18% for the mid-fed, and a 19% decrease for the long-fed market class with this revised factor applied.

Manure management emissions also have a high degree of uncertainty. Estimated manure emissions and mean results calculated using a Monte Carlo analysis of uncertainty are shown in Table 10. Calculated totals were slightly different than the mean results from the Monte Carlo analysis, because the uncertainty analysis applied skewed distributions for some factors such as nitrous oxide from land application of manure, and emissions of indirect nitrous oxide from ammonia volatilisation.

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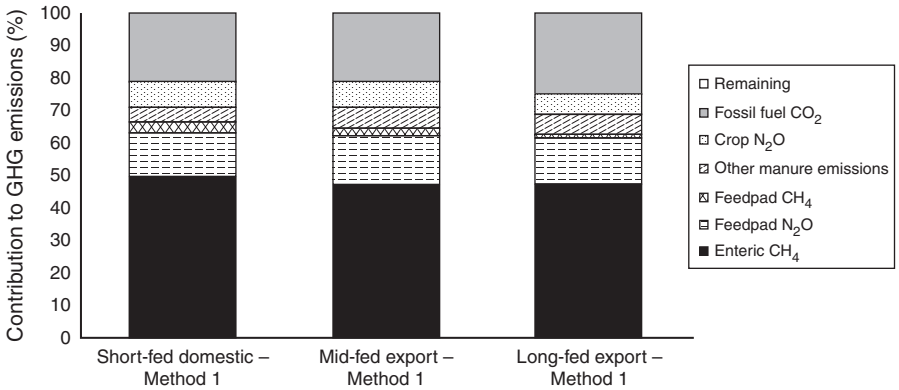

Fig. 4. Major sources of greenhouse gas per kg of liveweight gain from domestic short-fed, and mid-fed and long-fed export feedlot steers.

Mean emissions from LU and dLUC sources differed depending on the method applied. Where national estimates of LU and dLUC were applied, emissions from this source were negligible. Where state-based estimates of dLUC and high estimates of carbon loss from LU were applied, emissions were 1.19, 1.32 and 1.8 kg CO_2-e/kg LWG for the short-fed, mid-fed and long-fed cattle respectively, with differences associated with feed conversion ratio and grain production region.

Resources and GHG – cradle-to-farm gate

Liveweight gain during the feedlot finishing phase represents 26%, 35% and 44% of the total beef production from the primary production system, for the short-fed domestic, mid-fed export and long-fed export respectively.

However, the feedlot finishing phase contributed 34–72% of total energy demand, 80–96% of total crop land occupation, and 38–53% of fresh water consumption (see Fig. 5). Total energy demand was 10.0, 11.6 and 18.1 MJ per kilogram of LW finished for the short-fed domestic, mid-fed export and long-fed export respectively. Fresh water consumption was 296, 308 and 206 L, water stress was 99, 118 and 18 L H_2O-e, total land occupation was 106, 97 and 101 m^2/kg LW, and crop land occupation was 5.6, 8.9 and 24 m^2/kg LW finished for the short-fed domestic, mid-fed export and long-fed export, respectively. The feedlot finishing phase was found to contribute a relatively smaller proportion of overall impacts for total land occupation, at just 4%, 8% and 23% for the short-fed domestic, mid-fed export and long-fed export, respectively.

Greenhouse gas emissions were found to contribute a relatively smaller proportion of overall impacts (Fig. 5). Excluding LU and dLUC, GHG emissions were higher (9.9, 9.4 and 10.6 kg CO_2-e/kg LW finished for the short-fed domestic, mid-fed export and long-fed export, respectively), compared with the feedlot stage only. With LU and dLUC emissions included, results varied depending on the assumptions used. Where modest levels of vegetation carbon sequestration were included in the breeding and backgrounding stage (after Henry *et al*. 2015), LU and dLUC emissions were negligible for all market types. In this scenario, sequestration in the breeding and backgrounding stages compensated for emissions from crop land associated with feedlot ration production. Where some emissions from vegetation management were included in the breeding and backgrounding stages (after Henry *et al*. 2015), LU and dLUC emissions were 0.2–0.3 kg CO_2-e or 1.5–2.0 kg CO_2-e/kg LW finished depending on the method applied for estimating crop land emissions.

Table 10. Mean livestock manure greenhouse gas emissions per 100 kg of liveweight gain for two feedlots located in southern Queensland and southern New South Wales

	Queensland	Southern New South Wales
Excreted manure nitrogen	7.3	9.9
Excreted manure volatile solids	124.4	92.7
Emission source		
Feed pad nitrous oxide (kg CO_2-e)	68.8	92.7
Feed pad methane (kg CO_2-e)	11.7	2.9
Stockpile nitrous oxide (kg CO_2-e)	6.2	8.3
Stockpile methane (kg CO_2-e)	3.8	2.8
Indirect nitrous oxide (kg CO_2-e)	5.2	7.0
Pond methane (kg CO_2-e)	6.0	4.4
Total emissions (kg CO_2-e)	101.7	118.1
Mean emissions – uncertainty analysis (kg CO_2-e)	110.3 ± 54.7	127.8 ± 52.5

Discussion

The number of cattle finished on grain in Australia has increased significantly over the past 30 years. The net impact of this change has been a reduction in GHG emissions across the Australian herd (Wiedemann *et al*. 2015), in response to a shift from higher emission intensity grass-finished cattle, to lower emission intensity grain-finished cattle. However, increased production intensity via lot feeding has contributed to higher energy use (Wiedemann *et al*. 2015) and the impact on water use has not been studied in detail. This study is the first to provide a detailed analysis of resource use and GHG emissions of the grain finishing component of the supply chain for different market types. In the grain finishing stage, we found the feed conversion ratio, a major productivity factor for the industry, explained a high proportion of the variability in GHG emissions intensity, energy demand and arable land occupation,

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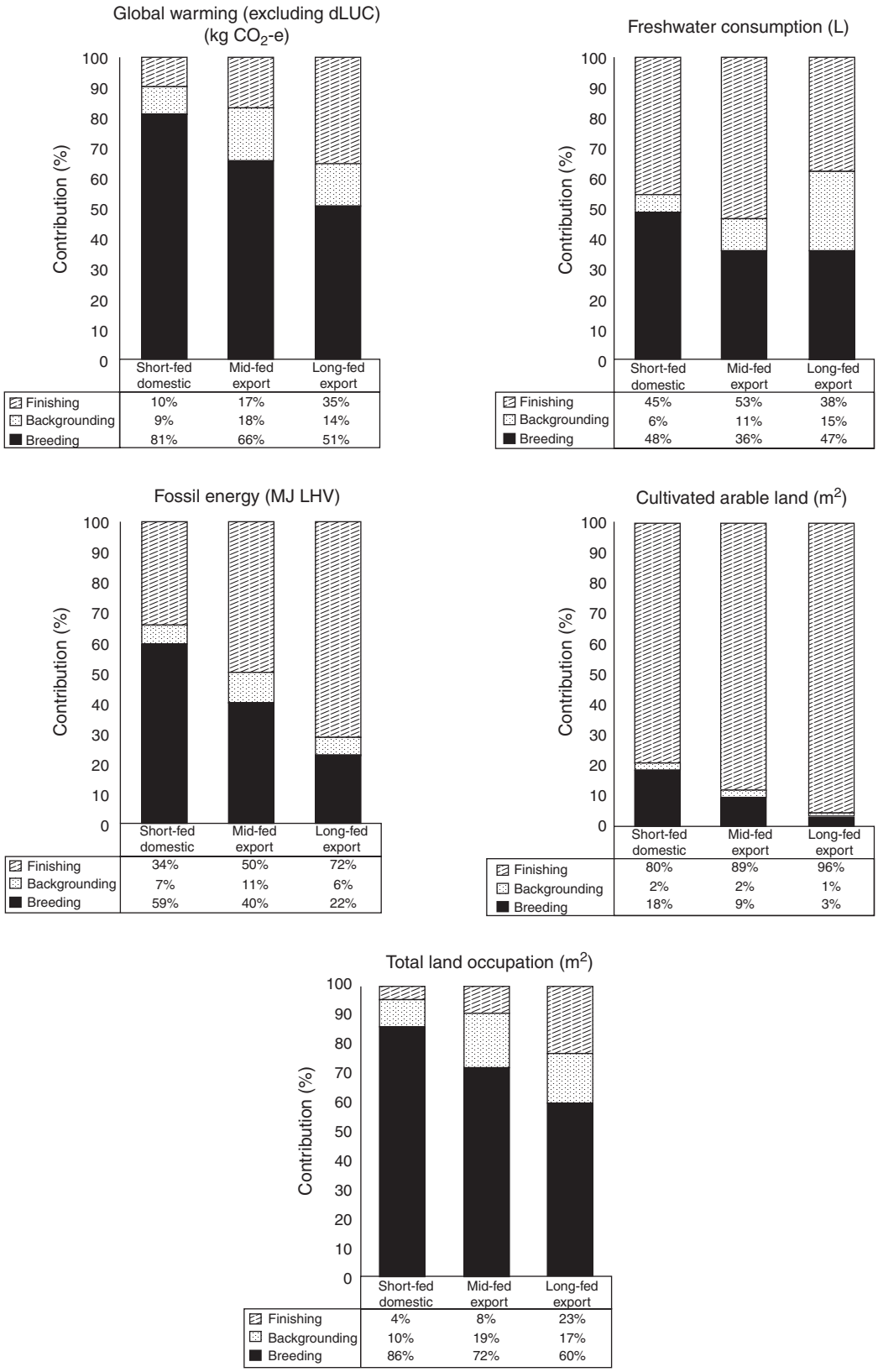

Fig. 5. Contribution of different stages of the supply chain to greenhouse gas emissions and resource use for grain-finished cattle.

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but not fresh water consumption or stress weighted water use. A significant but lesser association was also found between average daily gain and resource use and emissions intensity (results not shown). Fresh water consumption and stress-weighted water use were more heavily influenced by the use of irrigation for particular feed types such as cotton seed and regional water stress, rather than differences in livestock productivity.

Resource use – feedlot gate-to-gate

Water is a critical input for livestock production. A detailed analysis of water use at Australian feedlots by Davis *et al*. (2010*a*) showed that water use within the feedlot complex is dominated by cattle drinking water, with smaller contributions from feed management, cattle washing and other minor uses. In the present study, we used this dataset but expanded the analysis to include supply losses and irrigation water use both on-site and off-site. The fresh water consumption results reported by Davis *et al*. (2010*a*) converted to a LWG basis ranged from 22.5 to 32 L/kg LWG, which represented only ~10% of the fresh water consumption when irrigation and supply losses were included in the analysis (Fig. 1).

We found irrigation (including supply losses) to be the largest source of fresh water consumption, with mean contributions across the market types of between 57% and 91% for the feedlot finishing stage. Specifically, large volumes of water were associated with the production of cotton seed and cotton seed meal, which was a common feed input (~6% of ration) for the short-fed and mid-fed market types (see Table 5) and alone accounted for 25–35% of total fresh water consumption for these market types. In addition, the irrigation of hay and silage on feedlot farms consumed large volumes of water. This study found mean on-site irrigation (including supply losses) to be 49% of total fresh water consumption across all feedlots and market types. Among individual feedlots the contribution of irrigation water ranged from zero to 898.8 L/kg LWG in response to local irrigation water availability, which varied from feedlot to feedlot and was not strongly related to market type. Because of the wide range in irrigation water use between feedlots, no association was found between fresh water consumption and productivity for each market type. As only one feedlot was used to assess the long-fed market type, the total quantum, and contribution to water use may not be representative of this market type and these results should be viewed with caution.

Fossil fuel energy demand was dominated by feed production and milling rather than feedlot operations. When analysed separately, feed milling contributed 12% to overall energy demand whereas other feedlot operations contributed 11%. Although milling and general feedlot operations are the only parts of the operation under the direct control of feedlot operators, the total energy demand associated with production is more heavily influenced by feed conversion ratio because of the indirect influence of energy associated with feed production and transport. Total energy demand was considerably higher than reported by Davis *et al*. (2010*b*), who reported direct energy demand of 2.1–5.7 MJ for feedlot operations, but excluded energy demand associated with feed production.

Land occupation was dominated by feed production, whereas the occupation for the feedlot itself was insignificant because of the very high density of livestock on relatively small land areas.

GHG emissions – feedlot gate-to-gate

The GHG from sources other than LU and dLUC were dominated by enteric methane and emissions from manure, with smaller contributions from nitrous oxide from cropping and fossil fuel-related CO_2 (Fig. 4). Emissions intensity was largely governed by productivity factors, which influenced enteric methane, manure and feed-related emissions. The significance of ADG on emissions intensity was shown by Hunter and Niethe (2009) and Wiedemann *et al*. (2016*a*) and was confirmed in the present study (results not shown), though stronger regression relationships were found to feed conversion ratio here. Impacts from LU and dLUC were substantially different depending on the method applied, resulting in negligible change, or a substantial increase in GHG if LU soil carbon losses were assumed to be high. When the high estimates from these emission sources were included in the contribution analysis, impacts associated with feed production, transport and milling increased to between 38% and 42% of emissions in the gate-to-gate analysis. Considering the potential significance of LU and dLUC emissions, further research is required to improve this analysis for feed grain inventories in Australia.

Sensitivity to emission factors for GHG prediction

Emission intensity prediction is sensitive to specific emission estimation methods. The method used to calculate enteric methane emissions in the Australian GHG inventory predicts significantly higher emissions than the international default IPCC method (Dong *et al*. 2006). However, the Australian method has been found to predict emissions that are in closer agreement with the measurements made by McGinn *et al*. (2008) at one Qld feedlot, and are thus considered a robust representation of Australian feedlot production. We note that the Australian method predicts emissions (relative to GEI – 4.9–5.3%) that are mid-way between the recommended IPCC factors of 3% GEI for ~90% concentrate diets and 6.5% for diets with less than ~90% concentrates. For the domestic and mid-fed market types, the Australian method is therefore more conservative than the IPCC approach, though this is not the case for the long-fed market type. Further research would therefore be warranted to ascertain the level of agreement between predicted emissions using the preferred Australian inventory method, and measured enteric methane from long-fed cattle with lower concentrate diets. We found predicted nitrous oxide emissions from manure deposited to the feed pad to be a sensitive model parameter. Recent Australian research (Redding *et al*. 2015) demonstrated that emissions may be considerably lower from this source than those predicted using the NIR factors, and could result in total emissions per kilogram of beef being 16–19% lower than reported here. Further Australian research is required to confirm nitrous oxide from this source.

Cradle to farm gate

Feedlot finishing of beef cattle produces a substantial productivity increase compared with the grass-fed stages

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before the feedlot. Daily growth rates in the present study were ~3 times higher in the finishing phase compared with backgrounding or finishing on pasture, and finished weights are consequently achieved at a younger age than comparative grass-finished cattle. Compared with the companion analysis of grass-finished beef by Wiedemann *et al*. (2016*a*), GHG emissions excluding LU and dLUC were 15%, 23% and 13% lower for the domestic, mid-fed and long-fed export cattle compared with grass-fed cattle from similar herds. The lower emission intensity is associated with faster growth rates and therefore reduced age at slaughter for the grain finished cattle, resulting in lower lifetime enteric methane and manure emissions. Although emissions associated with fossil fuel energy use and crop production are higher for feedlot cattle, the reduced enteric methane emissions more than compensate for this increase. Similar results have been found by Peters *et al*. (2010*a*) and Pelletier *et al*. (2010) for cattle finished in feedlots in Australia and the USA. The impact of including LU and dLUC emissions on this result was quite dependent on the method applied. Using national inventory data where carbon sequestration in existing crop land is reported to be high, there were negligible impacts from LU and dLUC. However, where crop land was assumed to be a net source of emissions, this was a more significant source. Emissions from LU and dLUC sources should be interpreted with caution, because of the high degree of uncertainty in the estimation techniques (Henry *et al*. 2015). Further research is warranted to understand the magnitude of these emissions in Australian crop lands.

In contrast to the GHG results, we found fresh water consumption to be 49–51% higher and energy demand to be 24–38% higher for grain-finished cattle in the short-fed domestic and mid-fed export market classes. Crop land occupation and stress-weighted water use were also higher than grass-finished cattle from the same region (Wiedemann *et al*. 2016*a*). However, total land occupation was 14–24% lower, than grass-finished cattle from the same region. Although the magnitude of impacts per kilogram of LWG in the feedlot were quite different to impacts per kg of LW finished (cradle to gate), the differences between market types were similar for fresh water consumption, energy demand and crop land occupation. However, GHG emissions from the mid-fed export market type were lower than the short-fed domestic market type when the full supply chain impacts were considered, because of the comparatively larger reduction in slaughter age for these cattle and higher finished weights. This suggests that mid-fed cattle with finishing weights of ~650 kg LW are optimal from the perspective of emissions intensity.

Conclusions

Grain finishing was found to have relatively high energy demand, fresh water consumption and crop land occupation, whereas GHG emissions were lower than comparative grass-finished cattle. Emissions intensity was lowest for cattle finished on grain for 125 days to 650 kg LW, suggesting this market type is closest to optimum. The results suggest that increasing turnoff weight for the domestic market type would be one approach to improve the environmental efficiency of beef production in Australia. With increasing numbers of cattle finished on grain in Australia, these results suggest lower GHG emissions from cattle produced using these systems, but higher resource use in response to intensification. Further research is required to refine prediction methods for manure emissions, enteric methane in long-fed cattle and emissions from LU and dLUC for different parts of Australia's crop land. Variation in GHG, energy demand and crop land was largely explained by differences in feed conversion ratio, suggesting that this performance measure is an important indicator of environmental performance. With all other factors remaining equal, continued industry focus on improving feed conversion ratio is expected to yield production benefits and environmental improvements over time in the feedlot industry.

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